

**RKDF University Bhopal**  
**Faculty Profile**

| Basic Information                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                |                                                                                           |                                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------|
| Name                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | DR RITESH SADIWALA                                             |                                                                                           |                                    |
| Date of Birth                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 08.12.1978                                                     |                                                                                           |                                    |
| Designation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | PROFESSOR AND HEAD                                             |                                                                                           |                                    |
| Department                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | ELECTRONICS AND COMMUNICATION ENGINEERING                      |                                                                                           |                                    |
| Experience                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 22 YEARS                                                       |                                                                                           |                                    |
| Email ID                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <a href="mailto:ritesh14ci@gmail.com">ritesh14ci@gmail.com</a> |                                                                                           |                                    |
| Contact No                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 7049589573                                                     |                                                                                           |                                    |
| Educational Qualification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                |                                                                                           |                                    |
| Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Year                                                           | University                                                                                |                                    |
| UG                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2001                                                           | Rajiv Gandhi Proudlyogiki Vishwavidyalaya , Bhopal (G.E.C. Bhopal) at present UIT, Bhopal |                                    |
| PG                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2006                                                           | Maulana Azad National Institute of Technology, Bhopal                                     |                                    |
| M.Phil                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                |                                                                                           |                                    |
| Ph.D.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2017                                                           | RKDF University, Gandhinagar, Bhopal                                                      |                                    |
| Post Doctorate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                |                                                                                           |                                    |
| NET Qualified                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                |                                                                                           |                                    |
| Publications                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                |                                                                                           |                                    |
| No. of Research Papers Published(Attach Annexure of Research papers published)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 60 (Annexure A attached)                                       |                                                                                           |                                    |
| No.of Books Published                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 1 (Annexure B attached)                                        |                                                                                           |                                    |
| Books Chapters Published(Provide print of 1 <sup>st</sup> Page of Book)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 4 (Annexure B attached)                                        |                                                                                           |                                    |
| Research Scholars Supervised                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                |                                                                                           |                                    |
| Research Program                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Award                                                          | Under Supevision                                                                          | Name of the University             |
| Ph.D. (Annexure C attached)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 9                                                              | 1                                                                                         | RKDF UNIVERSITY,GANDHINAGAR,BHOPAL |
| M.Phil                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                |                                                                                           |                                    |
| PG Thesis /Dissertation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 12                                                             | 2                                                                                         | RKDF UNIVERSITY,GANDHINAGAR,BHOPAL |
| Area of Expertise(100 words)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                |                                                                                           |                                    |
| <p>My field is Electronics and Communication Engineering. My area of expertise consists of communication systems and in that particular wireless communication, optical communication and Network communication. I am learning and specializing in data transmission over computer networks, including protocols, network design, and the management of local and wide area networks (LANs and WANs) and trying to update my knowledge in it. I am also specialized in Transistor Design and Fabrication and in the development and deployment of sensor networks that collect data and have expertise in designing energy-efficient circuits and communication systems that are suitable for low-power wireless sensor applications.</p> |                                                                |                                                                                           |                                    |
| Award and Achievement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                |                                                                                           |                                    |
| Name of Award                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Description(With certified copy of the award)                  |                                                                                           |                                    |
| National                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                |                                                                                           |                                    |
| International                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                |                                                                                           |                                    |
| Conference/Seminar/Workshops/FDP                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                |                                                                                           |                                    |
| Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | No.                                                            |                                                                                           |                                    |
| Conference Attended                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 19                                                             |                                                                                           |                                    |
| Conference Organized                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 4                                                              |                                                                                           |                                    |
| Seminar Attended                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 7                                                              |                                                                                           |                                    |
| Seminar Organized                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2                                                              |                                                                                           |                                    |
| Workshop Attended                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 9                                                              |                                                                                           |                                    |
| Workshop Organized                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2                                                              |                                                                                           |                                    |
| FDP Attended                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2                                                              |                                                                                           |                                    |
| FDP Organized                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2                                                              |                                                                                           |                                    |
| Research Project                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                |                                                                                           |                                    |
| Name of Project                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Funding Agencies                                               | Amount                                                                                    |                                    |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                |                                                                                           |                                    |
| Any other Achievement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                |                                                                                           |                                    |
| Patents Granted/Published                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Title of the Patent                                            | Year                                                                                      |                                    |

|                          |                                                                                                          |      |
|--------------------------|----------------------------------------------------------------------------------------------------------|------|
| <b>Patents Granted</b>   | Smart Clinical Site Selecting Robotic Device                                                             | 2024 |
|                          | Smart Clinical Site Selecting Robotic Device                                                             | 2023 |
|                          | IOT Based Patient Monitoring System Using Sensors to Detect, Analyse And Monitor Two Primary Vital Signs | 2022 |
|                          | Secure and Dynamic Key Management In Wireless Sensor Networks                                            | 2022 |
|                          | Method for Migration of Sessions Among Cipher Machines Using Lagrange Interpolation                      | 2022 |
|                          | Touch Screen Devices Employing Graphene Electrodes Field of Invention Computer Science                   | 2021 |
| <b>Patents Published</b> |                                                                                                          |      |

# ANNEXURE A

## Research Publications (60)

### (A) International Journal Publications (40)

1. Ritesh C Sadiwala “*Convergence Towards Next Generation Wireless Networks*” IOSR Journal of Electrical and Electronics Engineering (IOSR-JEEE) e-ISSN: 2278-1676 Volume 4, Issue 4 (Jan. - Feb. 2013), pp 01-13.
2. Ritesh C Sadiwala “*Energy Efficiency for Long Term Evolution Cellular Networks*”, International Journal of Engineering Research & Technology (IJERT) ISSN: 2278-0181 Vol. 2 Issue 10, October – 2013, pp 1968-1982.
3. Ritesh Sadiwala and Dr.Manisha Sharma “*SECURITY THREATS OF VOIP*” Journal of Innovative trends in Science, Pharmacy & Technology, Issue1, Vol-1(1), August 2014, pp 7-18.
4. Ritesh Sadiwala and Dr.Manisha Sharma “*PERFORMANCE ANALYSIS OF VOIP NETWORKS USING NETWORK SIMULATORS-A REVIEW*” International Journal of Computer Networking and Communication (IJCNC), Vol. 2, No. 3(August 2014) ,pp 1-15.
5. Ritesh Sadiwala and Dr. MinalSaxena “*Performance Evaluation of Next Generation Networks using OPNET Simulator*” Communications on Applied Electronics (CAE) – ISSN: 2394-4714 Foundation of Computer Science FCS, New York, USA Volume 2 – No.4, July 2015, pp 32-37
6. Ritesh Sadiwala and Dr. MinalSaxena, “*Transformation and Analysis of Wire line Networks to Wireless Networks Using OPNET Simulator*” International Journal of Broadband Cellular Communication Vol. 1: Issue 1, July 2015, pp 20-35
7. Ritesh Sadiwala and Dr. MinalSaxena, “*Performance Evaluation of Quality Parameters in VOIP and PSTN Systems*” IRACST – International Journal of Computer Networks and Wireless Communications (IJCNC), ISSN: 2250-3501 Vol.5, No 5, October 2015, pp 621-627
8. EnamurRahmanShamsi and Dr.Ritesh Sadiwala, “*Survey of Clustering based Application Specific Routing Protocols in Wireless Sensor Network,*” SHODH SANGAM – A RKDF University Journal of Science and Engineering, (ISSN No. 2581-5806), Feb. 2019, Vol-2, Issue-3, pp. 55–58.
9. EnamurRahmanShamsi and Dr.RiteshSadiwala, “*Energy Estimated Mobile Sink based Routing Protocol in Wireless Sensor Network,*” International Journal of Research & Technology (IJRT), Dec 2018, Vol 06, Issue 06, pp. 25–28.
10. Jayant D Dorave and Dr. Ritesh Sadiwala, “*Compatibility and Integration of Smart Dust with various frequencies of electromagnetic spectrum and physical systems*”, The Splendid Journals, June 2019, Vol 2, Issue 3, pp 1-6.
11. Nadia Sharif and Dr.RiteshSadiwala, “*A Methodical and Erudite Study on 5G Wireless Networks*”, SHODH SANGAM -- A RKDF University Journal of Science and Engineering (ISSN No. 2581-5806), March 2020, pp 52-61.
12. Nadia Sharif and Dr.RiteshSadiwala, “*Performance Analysis of QOS Parameters of 5G Wireless Communication Networks*”, Research Journal of Engineering Technology and Management (ISSN: 2582-0028), Volume 03, Issue 02, June-2020, pp 66-75.
13. SanjerJilani and Dr.RiteshSadiwala, “*A Methodical and Erudite Study on 5G Wireless Networks*”, SHODH SANGAM (ISSN No. 2581-5806), A RKDF University Journal of Science and Engineering (ISSN No. 2581-5806) ,Vol.-04, No.-02, April-2021, pp 34-40.
14. SanjerJilani and Dr.RiteshSadiwala, “*Performance Analysis of QOS Parameters of 5G Wireless Communication Networks*”, Research Journal of Engineering Technology and Medical Sciences (ISSN: 2582-6212), Special Issue 2 (April 2020), pp 97-109

15. Rajnish Mishra and Dr. Ritesh Sadiwala, "Triangular Potential Well Approximation in Scaled MOSFET for Low Frequency Noise Modeling and Characterization". Research Journal of Engineering Technology and Medical Sciences (ISSN: 2582-6212), Volume 03, Issue 03, Sep-2020, pp 41-51.
16. Rajnish Mishra and Dr. Ritesh Sadiwala, "Survey on Modelling and Characterization of MOSFET for Low Frequency Noise", SHODH SANGAM -- A RKDF University Journal of Science and Engineering (ISSN No. 2581-5806), Vol-03, No.06, Nov 2020, pp 109-115.
17. Ambresh Patel and Dr. Ritesh Sadiwala, "Performance Analysis of Various CMOS Logics for High Speed VLSI Circuits", Samriddhi –JPSET, Volume 15(1), Jan 2023. **(UGC Care Listed )**
18. Ambresh Patel and Dr. Ritesh Sadiwala, "Optimizing and Recuperating the Leakages in Low Voltage CMOS Circuits" Samriddhi –JPSET, Volume 14, Issue 2, 2022, pp 1-4. **(UGC Care Listed )**
19. Chandan Kumar Roy and Dr. Ritesh Sadiwala, "Smart Environment for IOT based Disease Diagnosis Healthcare Framework", YMER, Volume 21, Issue 11, Nov 2022, pg 446-458. **[Indexed in SCOPUS]**
20. Jayant Dorave and Dr. Ritesh Sadiwala, "Numerous Pursuer Drones Immobilizing Multiple Evader Drones by Switching Target Enabled Communication Method in Smart Dust-IoT Environment", Design Engineering, 2021, pp 9735-9748. **[Indexed in SCOPUS]**
21. Jayant Dorave and Dr. Ritesh Sadiwala, "A Secure Communication Protocol for Unmanned Aerial Vehicles Using IOT Protocols", Samriddhi –JPSET, Volume 14(4), Oct 2022. **(UGC Care Listed)**
22. Jayant Hande and Dr. Ritesh Sadiwala, "Data Security-Based Routing in MANETs Using Key Management Mechanism" SN Computer Science, Jan 2024, SN Computer Science, <https://doi.org/10.1007/s42979-023-02409-5> **[Indexed in SCOPUS]**
23. Jaishri Tiwari and Dr. Ritesh Sadiwala, "Personality prediction from Five-Factor Facial Traits using Deep learning" Journal of Integrated Science and Technology: Vol. 11 No. 4 (2023) **[Indexed in SCOPUS]**
24. Hemant Rajoriya, Ritesh Sadiwala, "Deep compressive sensing and reconstruction algorithm in wireless Internet of Things", Journal of Integrated Science and Technology: Vol. 11 No. 2 (2023) **[Indexed in SCOPUS]**
25. Kuldeep Pandey, Ritesh Sadiwala, "Dual-Port MIMO antenna design for IoT: Analysis and Implementation, Journal of Integrated Science and Technology", Vol. 12 No. 3 (2024) **[Indexed in SCOPUS]**
26. Ajay Kumar Barapatre, Ritesh Sadiwala, "Energy and resources management for Multiple Access in Massive IOT network", Journal of Integrated Science and Technology: Vol. 11 No. 3 (2023) **[Indexed in SCOPUS]**
27. Jayant Y. Hande, Ritesh Sadiwala, "Optimization of energy consumption and routing in MANET using Artificial Neural Network", Journal of Integrated Science and Technology", Vol. 12 No. 1 (2024) **[Indexed in SCOPUS]**
28. Chandan Kumar Roy, Ritesh Sadiwala, "Smart internet of things (IoT) based healthcare framework environment for Chikungunya disease diagnosis", Journal of Integrated Science and Technology: Vol. 12 No. 4 (2024) **[Indexed in SCOPUS]**
29. Chandan Kumar Roy, Ritesh Sadiwala, "Smart environment for IoT based disease diagnosis healthcare framework", Journal of Integrated Science and Technology: Vol. 12 No. 2 (2024) **[Indexed in SCOPUS]**
30. Kuldeep Pandey, Ritesh Sadiwala, "Performance evaluation of FR2-based IoT and AI applications using a Quad Port Multiple Input Multiple Output Antenna", Journal of Integrated Science and Technology: Vol. 11 No. 4 (2023) **[Indexed in SCOPUS]**

31. Ajay K Barapatre, Ritesh Sadiwala, "Resource Allocation in Massive Internet of Things-Edge Network with Optimal Path Planning and Scheduling", Samriddhi: A Journal of Physical Sciences, Engineering and Technology, Volume 14, Issue 4, 2022, Print ISSN: 2229-7111, Online ISSN: 2454-5767, Doi: 10.18090/samriddhi.v14i04.04 **(UGC Care Listed)**
32. Jaishri Tiwari and Dr. Ritesh Sadiwala, "Personality Prediction from Handwriting using Fine-tuned Transfer Learning Models", SAMRIDDHI, ISSN: 2454-5767, Volume 15, Issue 1, 2023 **(UGC Care Listed)**
33. Kuldeep Pandey and Dr. Ritesh Sadiwala, "Parametric Analysis of Circular Patch Wideband Antenna for IOT Application", Lecture Notes in Electrical Engineering (Advances in VLSI, Communication, and Signal Processing), Oct 5-6 2021 & IIT BHU Varanasi. **[Indexed in SCOPUS]**
34. Hemant Rajoriya and Dr. Ritesh Sadiwala, "Deep compressive sensing and reconstruction algorithm in wireless Internet of Things" Journal of Integrated Science & Technology, ISSN: 2321-4635, J. Integr. Sci. Technol., 2023, 11(2), 487 URN:NBN: sciencein. jist.2023.v11.487. **[Indexed in SCOPUS]**
35. Hemant Rajoriya and Dr. Ritesh Sadiwala, "Data Traffic Reduction in Wireless Internet of Things Using Deep Compressive Sensing and Reconstruction Algorithm" A Journal of Physical Sciences, Engineering and Technology, Volume 15, Issue 1 (2023) Print ISSN: 2229-7111 Online ISSN: 2454-5767. **(UGC Care Listed)**
36. Jayant Y. Hande, and Ritesh Sadiwala, "Optimization of energy consumption and routing in MANET using Artificial Neural Network", Journal of Integrated Science & Technology Cite as: J. Integr. Sci. Technol., 2024, 12(1), 718. Published on: 11-Sep-2023 **(UGC Care Listed)**
37. Jayant Y. Hande and Dr. Ritesh Sadiwala, "A Secure Adaptive Bacterial Foraging Optimization Routing Algorithm for Mobile Ad Hoc Networks", TELEMATIQUE, Volume 21 Issue 1, 2022 **[Indexed in SCOPUS]**
38. Farooq Fareed Nabi, Dr Ritesh Sadiwala and Dr Ratnesh Jain, "Empirical Study: Status of Management of Microwave Radiation Hazards in the Government Organisation" Juni Khyat (□□□□ □□□□) ISSN: 2278-4632, Vol-14, Issue-4, No.04, April 2024 **(UGC Care Listed)**
39. Jayant Dorave and Dr. Ritesh Sadiwala, "IOT and Smart Dust Architecture Integration for Network Optimization using Artificial Neural Network", International Journal for Innovative Engineering and Management Research(IJIEMR), Vol-10, Issue 11, Nov 2021.
40. Jayant kumar Dorave and Dr. Ritesh Sadiwala, "Uplink and Downlink Variation in Drone Technology for Cloud, Edge, Fog and Smart Dust Integrated IoT Architecture: Demonstrated Over WSNs", AMSE 2021, Journal of Physics: Conference Series, IOP Publishing, doi:10.1088/1742-6596/2089/1/012023.

### ***(B)Conference Publications (20)***

1. Ritesh Sadiwala "Performance Evaluation of Quality Parameters for Integrated Unified Communication Network", 6<sup>th</sup> International Conference on "Computing for Sustainable Global Development", 13<sup>th</sup> - 15<sup>th</sup> March, 2019 Bharati Vidyapeeth's Institute of Computer Applications and Management (BVICAM), INDIA Com-2019, New Delhi (INDIA) IEEE Conference ID: 46181 **[Indexed in SCOPUS]**
2. Braj Kishor, Dr. Ritesh Sadiwala, "Dynamic CMOS Logic using clock gating in Deep Submicron Technology" 2<sup>nd</sup> International Conference on Contemporary Technological Solutions towards fulfillment of Social Needs, ShodhSangam September 28-29, 2019.
3. Braj Kishor, Dr. Ritesh Sadiwala "Performance analysis of Dynamic CMOS Logic using Submicron Technology" 3<sup>rd</sup> International Conference on Recent Advances in smart Innovative Ideas with Multidisciplinary Research, Nov 30- Dec 01, 2019.
4. Thirupathi Durgam and Dr. Ritesh Sadiwala, "Dynamic time assisted authentication protocol (DTAAP) for client-server in WSN-IoT environment", 2021 International Conference on Emerging Smart Computing and Informatics, ESCI 2021, 2021, pp. 583–586, ID no. 9397039 **[Indexed in IEEE Xplore, SCOPUS]**
5. Chandan Kumar Roy and Dr. Ritesh Sadiwala "Cloud-fog based healthcare framework to identify and prevent dengue fever outbreak", 2021 International Conference on Emerging Smart Computing and Informatics, ESCI 2021, 2021, pp. 537–541, ID no. 9396905 **[Indexed in IEEE Xplore, SCOPUS]**
6. Ambresh Patel and Dr. Ritesh Sadiwala "Low voltage cmos switch capacitor circuit", Proceedings of the 7<sup>th</sup> International Conference on Electrical Energy Systems, ICEES 2021, 2021, pp. 629–634, ID no. 9383686 **[Indexed in IEEE Xplore, SCOPUS]**

7. Chandan Kumar Roy and Dr. Ritesh Sadiwala “An Intelligent Fog-IoT based Disease Diagnosis Healthcare System”, 4<sup>th</sup> International Conference on Soft Computing and Signal Processing (ICSCSP-2021), Proceedings of ICSCSP-2021, Springer- Advances in Intelligent Systems and Computing (AISC) Series. **[Indexed in IEEE Xplore, SCOPUS]**
8. Thirupathi Durgam and Dr. Ritesh Sadiwala, “An Efficient Group Key Establishment for Secure Communication to multicast groups for WSN-IoT nodes “;4<sup>th</sup> International Conference on Soft Computing and Signal Processing (ICSCSP-2021), Proceedings of ICSCSP-2021, Springer- Advances in Intelligent Systems and Computing (AISC) Series. **[Indexed in IEEE Xplore, SCOPUS]**
9. Jaishri Tiwari and Dr. Ritesh Sadiwala, “Analysis of Personality Traits of Handwriting with Comparison of Different Techniques of Personality Detection” 2022 IEEE International Conference on Current Development in Engineering and Technology (CCET), 23-24 December 2022, DOI: [10.1109/CCET56606.2022.10080717](https://doi.org/10.1109/CCET56606.2022.10080717) **[Indexed in IEEE Xplore, SCOPUS]**
10. Ambresh Patel and Dr. Ritesh Sadiwala, “Low Voltage CMOS Switch Capacitor Circuit, 2021 7th International Conference on Electrical Energy Systems (ICEES), 11-13 February 2021, DOI: [10.1109/ICEES51510.2021.9383686](https://doi.org/10.1109/ICEES51510.2021.9383686) **[Indexed in IEEE Xplore, SCOPUS]**
11. Ajay Kumar Barapatre, Ritesh Sadiwala, “A Study on Resource Allocation for Edge Computing in IOT Networks”, International E-Conference on Impact of Science, Engineering and Technology Advancements in Humanities, Management, Medical and Nursing Sciences” on 19 Nov 2022.
12. Ajay Kumar Barapatre, Ritesh Sadiwala, “Collision Avoidance for Multiple Access in Massive IOT”, MPCST sponsored National Conference on Contemporary Technological Solutions towards fulfillment of Social Needs at RKDF University, Gandhi Nagar ,Bhopal on 03-04 Nov 2023.
13. Jaishri Tiwari and Dr. Ritesh Sadiwala, “Personality Detection with AI using Deep learning approach ”, International Conference on Recent Advances in Science and Engineering , SAGE University, Bhopal, April 23, 2022
14. Jayant Y. Hande and Dr. Ritesh Sadiwala, “Routing Protocols applied in disaster management for future research using MANET: A review”, National Conference - Advances in Science, Engineering & Technology for Sustainable Development Goals, held on Dec. 06-08, 2022 at Madhyanchal Professional University, Bhopal (M.P.)
15. Jayant Y. Hande and Dr. Ritesh Sadiwala, “Study of various Routing Protocols for MANETs: - A prospect review study”, International Conference on Futuristic Research in Modeling of Dynamical Systems, (ICFRMDS-2022) held on 4th and 5th of May 2022, at CMR INSTITUTE OF TECHNOLOGY, BENGALURU.
16. Jayant Y. Hande and Dr. Ritesh Sadiwala, “Comparative study of Various Optimization techniques in MANET”, International Conference on Advance research Tools Technique for Multidisciplinary Research design organised by Eudoxia Research University, USA and Eudoxia Research Centre, India held on 9th and 10th of December 2022.
17. Jayant Dorave and Dr. Ritesh Sadiwala, “Uplink and Downlink Variation in Drone Technology for Cloud, Edge, Fog and Smart Dust Integrated IoT Architecture: Demonstrated over WSNs”, 1<sup>st</sup> International Conference on Applied Mathematics, Modeling and Simulation in Engineering (AMSE 2021) , Stanley College of Engineering and Technology, Hyderabad, India held on 15-16 September 2021.
18. Kuldeep Pandey and Dr. Ritesh Sadiwala, “Dual Port Multiple Input Multiple Output Antenna for 5G Communication Applications”, in the 2<sup>nd</sup> International Conference on Advances in Signal Processing and Communication Engineering (ICASPACE-2023) **[Indexed in IEEE Xplore, SCOPUS]**
19. Jayant Dorave and Dr. Ritesh Sadiwala, “ IOT and Smart Dust Architecture Integration for Network Optimization using Artificial Neural Network”, International Journal for Innovative Engineering and Management Research (IJIEMR), Vol-10, Issue 11, Nov 2021 **[Indexed in IEEE Xplore, SCOPUS]**
20. Hande, J.Y., Sadiwala, Ritesh. Data Security-Based Routing in MANETs Using Key Management Mechanism. SN COMPUT. SCI. 5, 155 (2024). <https://doi.org/10.1007/s42979-023-02409-5>. **[Indexed in SCOPUS]**

(C) Paper Presentation International/ National Conferences (06)

- 1) Ritesh Sadiwala “Performance Evaluation of Quality Parameters for Integrated Unified Communication Network”, 6 th International Conference on “Computing for Sustainable Global Development”, 13th - 15th March, 2019 Bharati Vidyapeeth's Institute of Computer Applications and Management (BVICAM), INDIA Com-2019, New Delhi (INDIA) IEEE Conference ID: 46181[**Indexed in SCOPUS**]
- 2) BrajKishor, Dr. Ritesh Sadiwala, “Dynamic CMOS Logic using clock gating in Deep Submicron Technology” 2nd International Conference on Contemporary Technological Solutions towards fulfillment of Social Needs, ShodhSangam September 28-29, 2019.
- 3) BrajKishor, Dr. Ritesh Sadiwala “Performance analysis of Dynamic CMOS Logic using Submicron Technology” 3<sup>rd</sup> International Conference on Recent Advances in smart Innovative Ideas with Multidisciplinary Research, Nov 30- Dec 01, 2019.
- 4) Thirupathi Durgam and Dr. Ritesh Sadiwala, ”Dynamic time assisted authentication protocol (DTAAP) for client-server in WSN-IoT environment”, 2021 International Conference on Emerging Smart Computing and Informatics, ESCI 2021, 2021, pp. 583–586, ID no. 9397039[**Indexed in IEEE Xplore, SCOPUS**]
- 5) ThirupathiDurgam and Dr. Ritesh Sadiwala, “An Efficient Group Key Establishment for Secure Communication to multicast groups for WSN-IoT nodes “, 4<sup>th</sup> International Conference on Soft Computing and Signal Processing (ICSCSP-2021), Proceedings of ICSCSP-2021, Springer- Advances in Intelligent Systems and Computing (AISC) Series. [**Indexed in IEEE Xplore, SCOPUS**]
- 6) Dr. Ritesh Sadiwala, “Dynamic CMOS Logic using clock gating in Deep Submicron Technology” 2nd International Conference on Contemporary Technological Solutions towards fulfillment of Social Needs, Shodh Sangam September 28-29, 2019.

## ANNEXURE B

### (A) Book Published:-

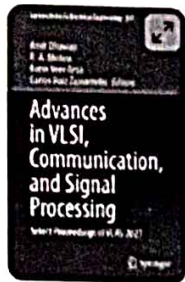
| S. No. | Title of Book/Monograph                                             | Authors             | Editor          | Year | ISBN No.      | Publisher Name  |
|--------|---------------------------------------------------------------------|---------------------|-----------------|------|---------------|-----------------|
| 1.     | Improvement Measures of Key Parameters for Future Wireless Networks | Dr. Ritesh Sadiwala | Ecaterina Focsa | 2022 | 9786202300377 | Scholars' Press |

### (B) Book Chapter Published (4)

- 1) Pandey, K., Shankhwar, A.K., Sadiwala, Ritesh (2022). Parametric Analysis of Circular Patch Wideband Antenna for IoT Application. In: Dhawan, A., Mishra, R.A., Arya, K.V., Zamarreño, C.R. (eds) Advances in VLSI, Communication, and Signal Processing. Lecture Notes in Electrical Engineering, vol 911. Springer, Singapore. [https://doi.org/10.1007/978-981-19-2631-0\\_9](https://doi.org/10.1007/978-981-19-2631-0_9) [Indexed in SCOPUS]
- 2) Durgam, T., Sadiwala, R. (2022). An Efficient Group Key Establishment for Secure Communication to Multicast Groups for WSN-IoT Nodes. In: Reddy, V.S., Prasad, V.K., Wang, J., Reddy, K. (eds) Soft Computing and Signal Processing. ICSCSP 2021. Advances in Intelligent Systems and Computing, vol 1413. Springer, Singapore. [https://doi.org/10.1007/978-981-16-7088-6\\_64](https://doi.org/10.1007/978-981-16-7088-6_64) [Indexed in SCOPUS]
- 3) Roy, C.K., Sadiwala, R. (2022). An Intelligent Fog-IoT-Based Disease Diagnosis Healthcare System. In: Reddy, V.S., Prasad, V.K., Wang, J., Reddy, K. (eds) Soft Computing and Signal Processing. ICSCSP 2021. Advances in Intelligent Systems and Computing, vol 1413. Springer, Singapore. [https://doi.org/10.1007/978-981-16-7088-6\\_33](https://doi.org/10.1007/978-981-16-7088-6_33) [Indexed in SCOPUS]
- 4) Pandey, K., Pandey, R., Sadiwala, R. (2024). Dual Port Multiple Input Multiple Output Antenna for 5G Communication Applications. In: Kumar Jain, P., Nath Singh, Y., Gollapalli, R.P., Singh, S.P. (eds) Advances in Signal Processing and Communication Engineering. ICASPACE 2023. Lecture Notes in Electrical Engineering, vol 1157. Springer, Singapore. [https://doi.org/10.1007/978-981-97-0562-7\\_36](https://doi.org/10.1007/978-981-97-0562-7_36) [Indexed in SCOPUS]

### Conference Proceedings (2)

1. Dr. Ritesh Sadiwala (2019) "Performance Evaluation of Quality Parameters for Integrated Unified Communication Network", IEEE Explorer, <https://ieeexplore.ieee.org/document/8991319> [Indexed in IEEE Explorer]
2. Ambresh Patel , Dr. Ritesh Sadiwala (2021) "Low Voltage CMOS Switch Capacitor Circuit" ,IEEE Explorer , <https://www.proceedings.com/content/058/058377webtoc.pdf> [Indexed in IEEE Explorer]

[Home](#) > Conference proceedings

# Advances in VLSI, Communication, and Signal Processing

Select Proceedings of VCAS 2021

| Conference proceedings | © 2022

## Overview

Editors: [Amit Dhawan](#), [R. A. Mishra](#), [Karm Veer Arya](#), [Carlos Ruiz Zamarreño](#)

Includes select papers from the International Conference on VLSI, Communication and Signal processing

Discusses current research in VLSI design, technology, and applications

Serves as a reference resource for researchers and practitioners in academia and industry

 Part of the book series: [Lecture Notes in Electrical Engineering \(LNEE, volume 911\)](#)

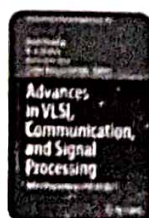
 46k Accesses

[Home](#) > [Advances in VLSI, Communication, and Signal Processing](#) > [Conference paper](#)

# Parametric Analysis of Circular Patch Wideband Antenna for IoT Application

| Conference paper | First Online: 05 October 2022

| pp 97–105 | [Cite this conference paper](#)



[Advances in VLSI, Communication,  
and Signal Processing](#)

[Kuldeep Pandey](#)  [A. K. Shankhwar](#) & [Ritesh Sadiwala](#)

 Part of the book series: [Lecture Notes in Electrical Engineering \(\(LNEE, volume 911\)\)](#)

 682 Accesses

## Abstract

In this paper, a novel topology microstrip dual-side printed antenna is presented for IoT-enabled applications. The novel structured antenna comprises a circular patch by cutting off the ground plane. The ground plane along with a small rectangular edge is exactly below the feed line to increase the antenna bandwidth and a circular notch on the surface of the ground plane is used to increase the antenna properties. Therewith, multiple rectangular notches in the ground plane have increased the effective area of the antenna. The proposed antenna covers the wideband frequency range of 1.15 to 4.83 GHz

[Home](#) > [Soft Computing and Signal Processing](#) > Conference paper

# An Efficient Group Key Establishment for Secure Communication to Multicast Groups for WSN-IoT Nodes

| Conference paper | First Online: 15 February 2022

| pp 705–714 | [Cite this conference paper](#)



[Soft Computing and Signal Processing](#)  
(ICSCSP 2021)

[Thirupathi Durgam](#) & [Ritesh Sadiwala](#)



Part of the book series: [Advances in Intelligent Systems and Computing](#)  
((AISC, volume 1413))



Included in the following conference series:  
[International Conference on Soft Computing and Signal Processing](#)

823 Accesses

## Abstract

Wireless sensor networks (WSNs) are an important element for Internet of Things technologies (IoT). The radio, multimedia, group communications deliver effective way

[Home](#) > [Soft Computing and Signal Processing](#) > Conference paper

# An Intelligent Fog-IoT-Based Disease Diagnosis Healthcare System

| Conference paper | First Online: 15 February 2022

| pp 359–370 | [Cite this conference paper](#)



[Soft Computing and Signal Processing](#)  
(ICSCSP 2021)

[Chandan Kumar Roy & Ritesh Sadiwala](#)

 Part of the book series: [Advances in Intelligent Systems and Computing](#)  
((AISC, volume 1413))

 Included in the following conference series:  
[International Conference on Soft Computing and Signal Processing](#)

 829 Accesses

## Abstract

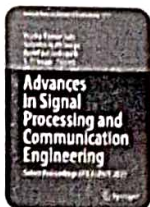
Modern health services are the greatest problem in developed countries in particular, where remote regions are not supplied with good quality drugs and hospitals. IoT is a major player in medical treatment to provide people with better clinical services, which

[Home](#) > [Advances in Signal Processing and Communication Engineering](#) > [Conference paper](#)

# Dual Port Multiple Input Multiple Output Antenna for 5G Communication Applications

| Conference paper | First Online: 04 July 2024

| pp 473–479 | [Cite this conference paper](#)



[Advances in Signal Processing and  
Communication Engineering](#)  
(ICASPACE 2023)

[Kuldeep Pandey](#) , [Rashmi Pandey](#) & [Ritesh Sadiwala](#)

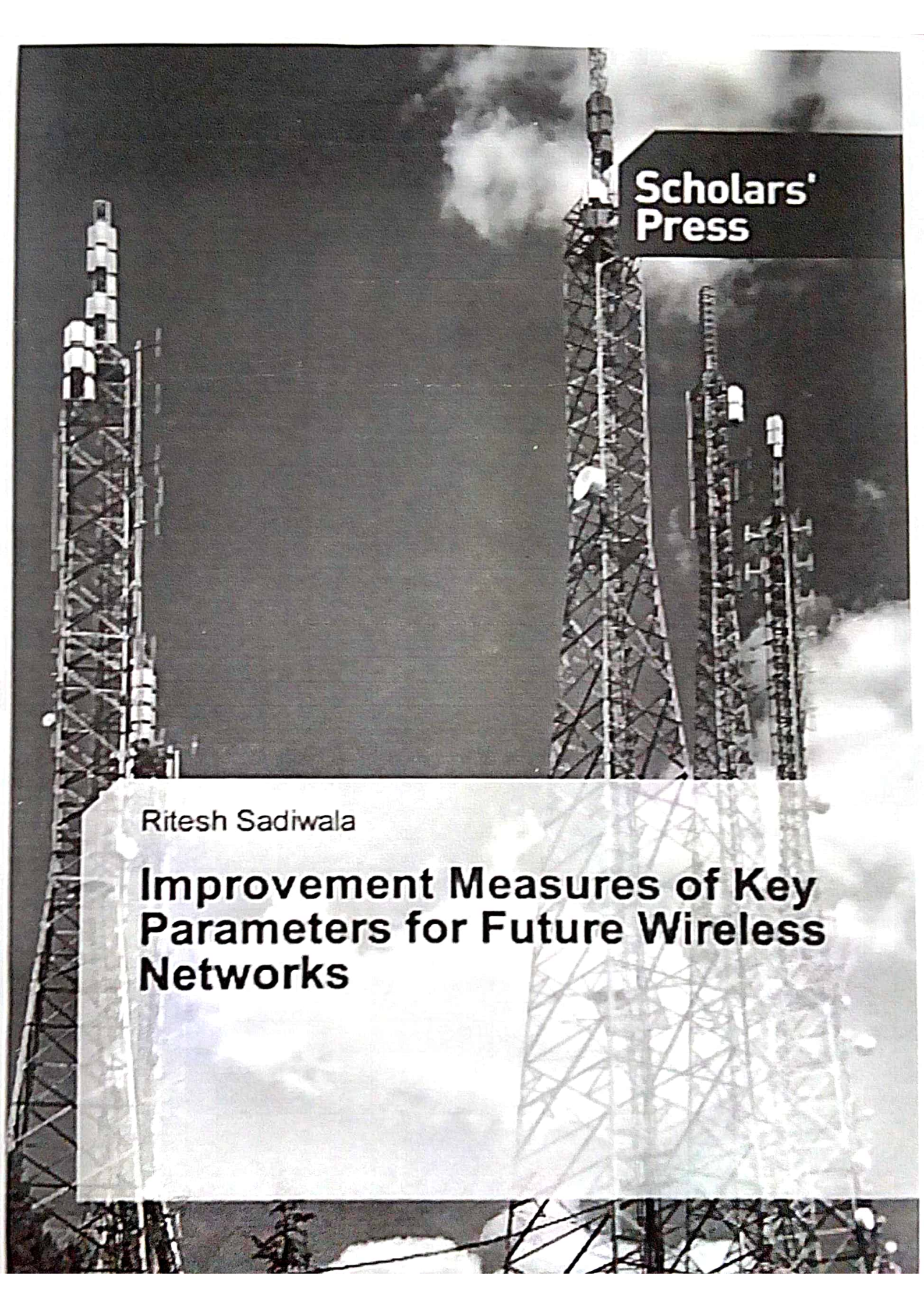
 Part of the book series: [Lecture Notes in Electrical Engineering](#) ((LNEE, volume 1157))

 Included in the following conference series:  
[International Conference on Advances in Signal Processing And Communication Engineering](#)

 114 Accesses

## Abstract

This paper presents the design of a two-element MIMO antenna for 5G applications. The dimensions of the proposed MIMO antenna remain at  $30 \text{ mm} \times 30 \text{ mm}^2$ , and it is constructed from an FR-4 substrate with a 1.6 mm thickness. A defective ground



**Scholars'  
Press**

Ritesh Sadiwala

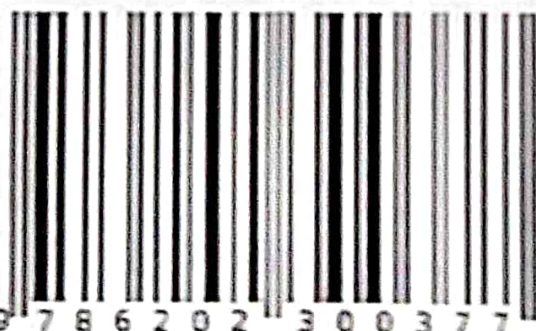
# **Improvement Measures of Key Parameters for Future Wireless Networks**

## Improvement Measures of Key Parameters for Future Wireless Networks

The enormous success of the Internet has led to a great revolution in Telecommunication. Today, Telecommunication has become an indispensable part of everybody's day-to-day activities. Various types of services like online video, online music, fast video streaming, video conferencing, VoIP, etc. are becoming a part of daily and routine activities. These applications demand high quality and congestion free networks particularly for voice and real time sessions. This can be possible when the key focus for telecommunication area will be on the Next Generation Wireless Networks (NGWNs). The idea behind this book is to Judge the key parameters of Next generation wireless networks so as to get a good network quality and a congestion free next generation network. The aim is achieved by analyzing various quality parameters like Jitter, Mean opinion Score(MOS), End to End Delay, etc. of the future networks. Next Generation Systems are advanced, fast and has several applications. The suggestions given in this book when implemented will definitely improve the performance of the Future Generation Networks by 90-95%.



Dr Ritesh Sadiwala has done his M.Tech from MANIT Bhopal (M.P.) India. He has done his Ph.D in Electronics and Communication Engineering. His areas of interest are Next Generation Networks, Wireless Sensor Networks. He has fifteen years of Industry and Eleven years of Teaching Experience. He has published many research papers in various Journals.



9 7 8 6 2 0 2 3 0 0 3 7 7

# ANNEXURE C

## Ph.D. Research Scholars Guidance

| Status    | Number |
|-----------|--------|
| Completed | 09     |
| On-going  | 01     |

### Details of the Ph.D Completed

| S.NO. | NAME OF THE STUDENT        | TITLE OF THESIS                                                                                                                            | YEAR OF AWARD |
|-------|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| 1.    | AJAY KUMAR BARAPATRE       | Resource Management for Multiple Access in Massive IoT Network                                                                             | 2024          |
| 2.    | JAISHRI TIWARI             | Image based Personality Prediction using Hybrid Deep Learning Approach                                                                     | 2024          |
| 3.    | JAYANT HANDE               | Optimization of energy consumption and routing in MANET using Artificial Neural Network                                                    | 2024          |
| 4.    | HEMANT RAJORIA             | Data Traffic Reduction in Wireless Internet of Things Using Deep Compressive Sensing and Reconstruction Algorithm                          | 2024          |
| 5.    | KULDEEP PANDEY             | Design and Performance Analysis of Multiple Input Multiple Output(MIMO) Smart Antenna for 5G                                               | 2024          |
| 6.    | AMBRESH PATEL              | Performance Analysis and Improvement Measures of Low Leakage CMOS Switch for Low Voltage Circuits                                          | 2024          |
| 7.    | CHANDAN KUMAR ROY          | Monitoring and Diagnosis of Some Human Diseases using IoT Systems                                                                          | 2025          |
| 8.    | FAROOQ FAREED NABI         | Microwave Radiation Induced Hazards Management in Government Organization Systems                                                          | 2025          |
| 9.    | JAYANTKUMAR DHANRAJ DORAVE | Analysis and Improvement of Performance Indicators at 2.4 GHz and 5GHz Network Communication Systems using Smart Dust Mote Sensor Networks | 2025          |